

CEO App Summary

One-page overview based on repo evidence from README and core backend/mobile files.

Repo gap: a custom local LLM that outperforms Codex/Claude is **Not found in repo**. The codebase shows a Gemini-powered assistant app.

What It Is / Who It Is For

CEO is a Jarvis-style personal AI assistant with a FastAPI backend and an Expo mobile/web client for voice or text chat.

Repo evidence shows **Gemini 2.0 Flash** as the model backend; a self-hosted local model is **Not found in repo**.

Primary persona: Charles ("Boss"), the named user in the system prompt and assistant copy.

General multi-user product positioning: Not found in repo.

How It Works

1. **Client:** `ChatScreen` captures typed text or hold-to-record audio; `SettingsScreen` stores the WebSocket URL in AsyncStorage.
2. **Transport:** `useWebSocket` opens an auto-reconnecting connection and sends JSON messages to FastAPI at `/ws`.
3. **Voice path:** voice uploads are base64-decoded, transcribed by `VoiceService` with `faster-whisper`, then forwarded as text.
4. **LLM path:** `GeminiService` sends the message to Gemini with automatic function calling enabled.
5. **Tools:** Gemini can call `ObsidianService`, `GitHubService`, and `ClaudeCodeService` (including git diff/commit/push helpers).
6. **Response path:** reply text is converted to MP3 by `edge-tts` and returned with the text payload; REST endpoints also expose `/health`, `/transcribe`, and `/speak`.

What It Does

- Accepts text chat and push-to-talk voice input from the mobile/web client.
- Auto-reconnects the client WebSocket and supports conversation reset.
- Transcribes speech on CPU with `faster-whisper`.
- Synthesizes spoken replies with `edge-tts`.
- Reads, writes, lists, and searches an Obsidian vault.
- Lists GitHub repos/issues and reads repo files when `GITHUB\_TOKEN` is set.
- Runs Claude Code CLI prompts and enforces a git safety gate before commit or push.

How To Run

1. **Backend:** in `backend/`, copy `.env.example` to `.env`, set `GEMINI\_API\_KEY`, install `ffmpeg`, then run `start.bat`.
 2. **Mobile/web:** in `mobile/`, run `npm install --legacy-peer-deps`, then `npx expo start` or `npx expo start --web`.
 3. **Connect:** in the app Settings screen, set `ws://<desktop-ip>:8000/ws`; remote access uses ngrok.
- Minimal prerequisites:** Python 3.10+, Node.js 18+, ffmpeg, and a Gemini API key.

Not Found In Repo

- A custom local LLM training or inference stack.
- Benchmarks or evidence that this app outperforms Codex or Claude.
- Apple Silicon / Mac mini 24/7 deployment instructions beyond generic local hosting notes.